Chapter 2. Distribution-Free Inference for One Population

Sign test, Wilcoxon signed-rank test, Hodges–Lehmann estimator, efficiency comparisons

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Setting

$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with median θ; test $H_0 : \theta = \theta_0$, estimate θ and build intervals, assuming F continuous (sign test) or continuous and symmetric about θ (signed-rank test).

2.1 The sign test

Motivation. For any continuous F the number of observations above the median is exactly binomial.

Definition (Sign statistic)

$S = S(\theta_0) = \sum_{i=1}^{n} I(X_i > \theta_0)$, the number of observations above θ_0.

Proposition 2.1

If $P(X_i = \theta_0) = 0$ and θ_0 is the median of F, then $S \sim B(n, \frac{1}{2})$ for every such F.

Proof. The indicators $I(X_i > \theta_0)$ are i.i.d. Bernoulli with success probability $P(X_1 > \theta_0) = 1 - F(\theta_0) = \frac{1}{2}$. ■

So the sign test is distribution-free. Under $\theta > \theta_0$, $p_1 = P(X > \theta_0) > \frac{1}{2}$: reject for large S, $p = P(S \geq s_{\text{obs}})$; two-sided $p = 2\min\{P(S \geq s_{\text{obs}}), P(S \leq s_{\text{obs}})\}$.

Proposition 2.2 (Rejection region)

The smallest c with $P(S \geq c) \leq \alpha$ under H_0 is $c = Q_B(1-\alpha) + 1$, where Q_B is the quantile function of $B(n, \frac{1}{2})$.

Proof. $P(S \geq c) \leq \alpha \iff F_B(c-1) \geq 1-\alpha$, and the smallest integer $c-1$ with this property is $Q_B(1-\alpha)$ (Proposition 1.1). ■

The attained size is usually below α (discreteness). For large n, $(S - n/2)/\sqrt{n/4} \dot{\sim} N(0,1)$.

Ties. Discard observations equal to θ_0 and reduce n; given the number discarded, the remaining signs are i.i.d. fair coins when $P(X < \theta_0) = P(X > \theta_0)$.

Example 2.1

Twelve paired differences (after − before) have 9 positive signs. For $H_1 : \theta > 0$, $p = P(S \geq 9) = (220 + 66 + 12 + 1)/4096 = 0.073$.

Power. Under an alternative $S \sim B(n, p_1)$, $p_1 = P(X > \theta_0)$, so the power is $1 - F_B(c-1; n, p_1)$; it depends on the shape of F.

Example 2.2 (Sign test vs. z-test)

Let $n = 20$, $\theta_0 = 0$, $\theta = 0.5$, standard deviation 1. The size-≤ 0.05 sign test rejects when $S \geq 15$ (actual size 0.021).

- Normal $N(0.5, 1)$: $p_1 = \Phi(0.5) = 0.69$, power 0.38; the z-test has power $\Phi(0.5\sqrt{20} -$

$1.645) = 0.72$.

- Laplace, same mean and variance (scale $1/\sqrt{2}$): $p_1 = 1 - \frac{1}{2}e^{-0.5\sqrt{2}} = 0.75$, power 0.63; z-test still about 0.72.

The sign test gains with heavy tails (Section 2.3).

2.1.1 Confidence interval for the median

Proposition 2.3

Let F be continuous, $X_{(1)} < \cdots < X_{(n)}$ the order statistics and $l = Q_B(\alpha/2)$, $u = n - l + 1$. Then

$$P(X_{(l)} \le \theta < X_{(u)}) = \sum_{k=l}^{n-l} \binom{n}{k} 2^{-n} \ge 1 - \alpha.$$

Proof. $X_{(l)} \le \theta$ iff at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ iff at least $n-u+1 = l$ do. So the event is $\{l \le S(\theta) \le n-l\}$ with $S(\theta) \sim B(n, \frac{1}{2})$. By symmetry $P(S \le l-1) = P(S \ge n-l+1)$, and each is $\le \alpha/2$ by the choice of l (if $P(S \le Q_B(\alpha/2)) = \alpha/2$ exactly, take $l = Q_B(\alpha/2) + 1$). ■

For $n = 15$, $\alpha = 0.05$: $l = 4$, and $(X_{(4)}, X_{(12)})$ has coverage 0.965. For large n, $S \dot{\sim} N(n/2, n/4)$ gives

$$l \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}, \qquad u \approx \frac{n}{2} + z_{\alpha/2}\frac{\sqrt{n}}{2} + 1 \qquad \text{(rounded to integers)},$$

an asymptotically valid distribution-free interval. As the level decreases it shrinks to the sample median.

For any quantile $Q(p)$: $\#\{X_i \le Q(p)\} \sim B(n, p)$, so $(X_{(l)}, X_{(u)})$ covers $Q(p)$ with probability $\sum_{k=l}^{u-1} \binom{n}{k} p^k (1-p)^{n-k}$.

2.2 The Wilcoxon signed-rank test

Motivation. Under symmetry the ranks of $|X_i|$ add information while keeping the test distribution-free.

Assume now that F is continuous and symmetric about θ. Let $D_i = X_i - \theta_0$, $V_i = I(D_i > 0)$, and let R_i be the rank of $|D_i|$ among $|D_1|, \ldots, |D_n|$.

Definition (Signed-rank statistic)

$SR_+ = \sum_{i=1}^{n} V_i R_i$, the sum of the ranks of the positive deviations.

Lemma 2.4

Under H_0 (symmetry about θ_0), $(V_1, \ldots, V_n)$ are i.i.d. Bernoulli($\frac{1}{2}$) and independent of $(|D_1|, \ldots, |D_n|)$. Consequently

$$SR_+ \overset{d}{=} \sum_{k=1}^{n} k B_k, \qquad B_k \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\tfrac{1}{2}).$$

Proof. For a symmetric continuous D, $\operatorname{sgn}(D)$ and $|D|$ are independent with $P(D > 0) = \frac{1}{2}$: $P(D > 0, |D| \le t) = P(0 < D \le t) = \frac{1}{2}P(|D| \le t)$. Independence across i gives the first claim.

Given the ranks, (R_i) is a permutation π of $(1,\dots,n)$ and $\sum_i V_i R_i = \sum_k k V_{\pi^{-1}(k)}$, with $(V_{\pi^{-1}(k)})_k$ again i.i.d. fair coins. ■

Proposition 2.5

Under H_0, the null distribution of SR_+ depends only on n, is symmetric about $n(n+1)/4$, and

$$E(SR_+) = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(SR_+) = \frac{n(n+1)(2n+1)}{24}, \qquad \frac{SR_+ - E(SR_+)}{\sqrt{\mathrm{Var}(SR_+)}} \xrightarrow{d} N(0,1).$$

Proof [Supplement]. From Lemma 2.4, $E = \frac{1}{2}\sum k$ and $\mathrm{Var} = \frac{1}{4}\sum k^2$. Replacing B_k by $1 - B_k$ (same law) maps SR_+ to $\frac{n(n+1)}{2} - SR_+$, giving symmetry. For normality use Lyapunov's CLT with $U_k = kB_k$: $\sum_k E|U_k - k/2|^3 = \sum k^3/8 = O(n^4)$, while $s_n^3 = \mathrm{Var}(SR_+)^{3/2} \asymp n^{9/2}$, so the Lyapunov ratio is $O(n^{-1/2}) \to 0$. ■

The exact null distribution enumerates the 2^n equally likely sign patterns (*sign-flip* principle): $p = P(SR_+ \geq sr_{\mathrm{obs}})$ against $\theta > \theta_0$, $P(SR_+ \leq sr_{\mathrm{obs}})$ against $\theta < \theta_0$, and twice the smaller one two-sided (the null law is symmetric).

Example 2.3

Deviations -0.3, 0.8, 1.5, 2.2, 2.9: absolute ranks $1,\dots,5$, only rank 1 negative, so $SR_+ = 14$. Of the 32 sign patterns, $SR_+ \geq 14$ for "all positive" (15) and "only rank 1 negative" (14): $p = 2/32 = 0.0625$. The sign test gives $P(S \geq 4) = 6/32 = 0.19$.

Ties. Drop zero deviations; give tied $|D_i|$ mid-ranks. The sign-flip distribution of the actual mid-ranks is still exact conditionally; for the normal approximation use $\mathrm{Var}(SR_+) = \frac{1}{4}\sum_i R_i^2$.

2.2.1 Walsh averages, the Hodges–Lehmann estimator and intervals

The $N = n(n+1)/2$ **Walsh averages** are $W_{ij} = (X_i + X_j)/2$, $1 \leq i \leq j \leq n$.

Proposition 2.6

If no $|X_k - \theta|$ are tied and no W_{ij} equals θ, then $SR_+(\theta) = \#\{(i \leq j) : W_{ij} > \theta\}$.

Proof [Supplement]. Let $d_k = X_k - \theta$, so $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i,j\}$ to the $k \in \{i,j\}$ with larger $|d_k|$; then $\mathrm{sgn}(d_i + d_j) = \mathrm{sgn}(d_k)$, and k receives the pairs $\{k,m\}$ with $|d_m| \leq |d_k|$, i.e. R_k pairs. Hence $\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k$. ■

So SR_+ is the sign statistic of the Walsh averages, and Proposition 2.3 transfers.

Corollary 2.7

Let $W_{(1)} < \cdots < W_{(N)}$ and $l = Q_{SR}(\alpha/2)$ (quantile of the null distribution of SR_+), $u = N - l + 1$. Then $P(W_{(l)} \leq \theta < W_{(u)}) \geq 1 - \alpha$. The point estimator obtained as the level shrinks is the **Hodges–Lehmann estimator** $\hat{\theta}_{HL} = \mathrm{median}_{i \leq j}\, W_{ij}$.

Proof. $\theta < W_{(a)} \iff SR_+(\theta) > N - a$ and $\theta \geq W_{(a)} \iff SR_+(\theta) \leq N - a$; at the true θ, $SR_+(\theta)$ has the null law, symmetric about $N/2$. Repeat the proof of Proposition 2.3. ■

Paired data: apply these methods to the differences.

2.3 Comparing the t-, sign and signed-rank procedures

Validity. The sign test is exact for continuous F, the signed-rank test for continuous symmetric F, the t-test only under normality. For asymmetric F the t-test does not test the median. Nominal 5% one-sided t-test:

- light tails (uniform): size slightly above 5% for $n \approx 5$, essentially correct by $n \approx 10$;
- infinite variance (slash, Cauchy): size well below 5%, not improving with n (CLT fails);
- skewed (exponential, testing the mean): slow convergence, size noticeably above 5% even at $n = 100$;
- skewed, testing the median with t: centered at the wrong value, meaningless.

Only valid tests (or ones with corrected critical values) are compared on power, which depends on F, n and α.

Efficiency. $RE(\hat\theta_1, \hat\theta_2) = \mathrm{MSE}(\hat\theta_2)/\mathrm{MSE}(\hat\theta_1)$; its limit is the **asymptotic relative efficiency** (ARE). For tests the (Pitman) ARE is the limiting ratio N_2/N_1 of sample sizes for equal power at $\theta_n \to \theta_0$; here it equals the ARE of $\bar X_n$, median X_i, $\hat\theta_{HL}$.

For intervals of equal coverage, efficiency is $1/\text{length}^2$. If $\hat\theta \dot\sim N(\theta, v/n)$, $\hat\theta \pm z_{\alpha/2}\sqrt{v/n}$ has length

$$2z_{\alpha/2}\sqrt{v/n} \approx 2z_{\alpha/2}\sqrt{\mathrm{MSE}(\hat\theta)}.$$

so interval ARE is the ratio of MSEs: estimators, tests and intervals rank alike. E.g. for a normal population $ARE(\bar X_n, \text{median } X_i) = \pi/2 \approx 1.57$.

Proposition 2.8

Let F have density f symmetric about θ, with $f(\theta) > 0$ and variance σ^2 (possibly infinite). Then

$$\sqrt{n}(\bar X_n - \theta) \xrightarrow{d} N(0, \sigma^2), \qquad \sqrt{n}(\text{median } X_i - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(\theta)^2}\Big),$$
$$\sqrt{n}(\hat\theta_{HL} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{12(\int f^2)^2}\Big).$$

Hence

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Proof for the median; sketch for HL. $P(\text{median } X_i \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$, the count being $B(n, p_n)$, $p_n = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$; the normal approximation gives $\Phi(2f(\theta)t)$, variance $1/(4f(\theta)^2)$. For $\hat\theta_{HL}$, by Proposition 2.6, $\hat\theta_{HL} \le \theta + t/\sqrt{n}$ iff $SR_+(\theta + t/\sqrt{n}) \le N/2$; this statistic has mean $N/2 - n^{3/2}t\int f^2 + o(n^{3/2})$ and variance $\sim n^3/12$, giving $\Phi(t\sqrt{12}\int f^2)$. ■

F	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	1/3	1	1/3
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.955$	2/3
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.10$	3/4
Laplace	2	1.5	4/3
Cauchy, slash	∞	∞	finite

(Check: $N(0,1)$ has $f(0)^2 = \frac{1}{2\pi}$, $\int \phi^2 = \frac{1}{2\sqrt{\pi}}$; Laplace $\frac{1}{2}e^{-|x|}$ has $\sigma^2 = 2$, $f(0) = \frac{1}{2}$, $\int f^2 = \frac{1}{4}$.) Over all symmetric F, $ARE(SR, t) \geq 0.864$ (Hodges–Lehmann): signed-rank costs little under normality and can gain a lot; the sign test wins only for very heavy tails.

2.4 Selected exercises

Motivation. A sign test for a non-median quantile, and the tie-corrected signed-rank moments used by software.

Exercise 2.4 (A sign test for the 10th percentile)

Ten foldable phones were folded until failure. The numbers of folds (in units of 10^4) were

$$7,\ 9,\ 12,\ 39,\ 28,\ 11,\ 22,\ 17,\ 14,\ 9.$$

Can one claim that at least 90% of such phones survive 10^5 folds? Use a sign-type test at level 0.1, and give a 90% confidence bound for the 10th percentile of the number of folds.

Solution.

Step 1. (Hypotheses.) Let $\xi = Q(0.1)$ of the continuous F. "At least 90% survive 10 units" means $P(X < 10) \leq 0.1$, i.e. $\xi \geq 10$; as the claim, it is the alternative: $H_0 : P(X < 10) = 0.1$ $(\xi = 10)$ vs. $H_1 : P(X < 10) < 0.1$ $(\xi > 10)$.

Step 2. (Test.) $K = \#\{i : X_i < 10\} \sim B(10, 0.1)$ under H_0 (Proposition 2.1 with $\frac{1}{2} \to 0.1$); small K supports H_1. Here $7, 9, 9$ give $k = 3$ and

$$p = P(K \leq 3) = 0.987.$$

H_0 is not rejected: three early failures exceed the one expected under H_0.

Step 3. (Confidence bound.) By Section 2.1.1, $J = \#\{i : X_i \leq \xi\} \sim B(10, 0.1)$, and (no ties a.s.) $\xi < X_{(u)} \iff J \leq u - 1$. Since $P(J \leq 1) = 0.736 < 0.9 \leq 0.930 = P(J \leq 2)$, take $u = 3$:

$$P(\xi < X_{(3)}) = 0.930 \geq 0.9.$$

$X_{(3)} = 9$ (the rounding tie $9, 9$ is harmless). A two-sided interval is impossible: $P(\xi < X_{(1)}) = 0.9^{10} = 0.349 > 0.05$.

With confidence $\geq 90\%$ the 10th percentile is below 9×10^4 folds: the claim is not supported. ■

Exercise 2.8 (Signed-rank moments with zeros and ties)

To test $H_0 : \theta = \theta_0$, let $D_i = X_i - \theta_0$. Rank all n values $|D_i|$ (zeros included, occupying the lowest ranks) with mid-ranks for ties; SR_+ is the rank sum of the positive D_i. Suppose t_0 of the D_i are zero and the nonzero $|D_i|$ form tie groups of sizes $t_1, \ldots, t_g$ (e.g. $D = 0, 0, 1, 1, 2, 3, 3, 3$: $n = 8$, $t_0 = 2$, sizes $2, 1, 3$). Show that under H_0 (symmetry of D about 0), conditionally on $|D_1|, \ldots, |D_n|$,

$$E(SR_+) = \frac{n(n+1) - t_0(t_0+1)}{4},$$
$$\mathrm{Var}(SR_+) = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{24} - \frac{\sum_l (t_l^3 - t_l)}{48}.$$

Solution.

Step 1. (Signs given magnitudes.) Symmetry gives, for $A \subset (0,\infty)$, $P(D>0, |D| \in A) = P(D<0, |D| \in A)$, and the two events partition $\{|D| \in A\}$; so $P(D>0 \mid |D|) = \frac{1}{2}$ on $\{|D|>0\}$. Given $(|D_j|)$, the $V_i = I(D_i>0)$, $i \in \mathcal{N} = \{i : D_i \neq 0\}$, are i.i.d. Bernoulli($\frac{1}{2}$), $V_i = 0$ otherwise, and the ranks are fixed: $SR_+ = \sum_{i\in\mathcal{N}} V_i R_i$.

Step 2. (Mean.) $E(SR_+ \mid |D|) = \frac{1}{2}\sum_{i\in\mathcal{N}} R_i$. Zeros take positions $1,\dots,t_0$, and mid-ranks preserve sums over tie groups, so

$$\sum_{i\in\mathcal{N}} R_i = \sum_{k=t_0+1}^{n} k = \frac{n(n+1) - t_0(t_0+1)}{2}.$$

Step 3. (A tie group.) A group in positions $a+1,\dots,a+t$ gets $\bar{r} = a + \frac{t+1}{2}$, and

$$\sum_{k=a+1}^{a+t} k^2 - t\bar{r}^2 = \sum_{k=a+1}^{a+t} (k-\bar{r})^2 = \sum_{j=1}^{t} \Big(j - \frac{t+1}{2}\Big)^2 = \frac{t(t^2-1)}{12}.$$

Step 4. (Variance.) $\mathrm{Var}(SR_+ \mid |D|) = \frac{1}{4}\sum_{i\in\mathcal{N}} R_i^2$, and by Step 3

$$\sum_{i\in\mathcal{N}} R_i^2 = \sum_{k=t_0+1}^{n} k^2 - \sum_{l} \frac{t_l^3 - t_l}{12} = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{6} - \sum_{l} \frac{t_l^3 - t_l}{12}.$$

Divide by 4. With $t_0 = 0$ and no ties this is Proposition 2.5.

Check: nonzero mid-ranks $3.5, 3.5, 5, 7, 7, 7$ have sum 33 and sum of squares 196.5, so $E = 16.5 = (72-6)/4$, $\mathrm{Var} = 49.125 = (1224-30)/24 - 30/48$. ■